Mohammad Yousaf

Forward Deployed Engineer | AI Systems | Full-Stack | Cloud & Infrastructure

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Professional Summary

Forward Deployed Engineer with hands-on experience building and deploying **AI-powered, full-stack, and infrastructure systems** for enterprise and government-related environments in the UAE. Experienced in translating operational requirements into working software, from AI assistants and optimization systems to real-time APIs, digital twins, secure container environments, and user-facing applications. Built systems processing **10K+ data points per session**, delivered enterprise AI features in short development cycles, and deployed solutions across Linux, Docker, Kubernetes, FastAPI, React, and cloud-oriented environments. Strong technical range across **AI/ML, backend engineering, infrastructure, security, optimization, and customer-facing technical delivery**.

Technical Skills

Software Engineering: Python, C, C++, Bash, JavaScript, TypeScript, React, Vite, Node.js, Next.js, REST APIs
Backend & AI Applications: FastAPI, LangChain, LangGraph, RAG, FAISS, LLMs, Multi-Agent Systems, Tool Use, AI Evaluation
AI/ML: PyTorch, TensorFlow, Scikit-learn, Computer Vision, Transformers, CNNs, NLP, Transfer Learning, ONNX, TensorRT
Infrastructure & Deployment: Docker, Kubernetes, AWS, Linux, Nginx, CI/CD, ESXi, Proxmox, RBAC, System Hardening
Data & Optimization: SQL, Pandas, NumPy, Spark, Hadoop, MariaDB/MySQL, OR-Tools, SimPy, Monte Carlo Simulation
MLOps & Observability: MLflow, Weights & Biases, Kubeflow, Airflow, Model Monitoring, Evaluation Pipelines

Professional Experience

AI Engineer – ADNOC (Contract) Jan 2026 – Jun 2026
Abu Dhabi, UAE

- Worked across AI, backend, and frontend layers to deliver capabilities for industrial digital twin and maintenance workflows.
- Engineered real-time inference pipelines processing **10K+ industrial data points per session** with **<200 ms latency**.
- Built streaming AI assistant APIs for maintenance workflows, reducing mean query resolution time by **35%**.
- Developed React/Vite interfaces for 3D industrial digital twin systems, contributing to an estimated **40% reduction in asset inspection time**.
- Prototyped and shipped **3 enterprise AI features** with cross-functional teams during 6-week delivery cycles.

Secure Environment Deployment – Ministry of Defence / 42 Abu Dhabi Oct 2025
Abu Dhabi, UAE

- Deployed isolated Linux container environments for secure technical training, reducing provisioning time by **60%**.
- Diagnosed and resolved **10+ deployment and environment issues** while supporting reliable delivery in a restricted infrastructure setting.
- Documented deployment outcomes and technical resolutions for senior stakeholders.

Systems & Lab Infrastructure Engineer – Khalifa Fund (MZN Hub) Jul 2025
Al Ain, UAE

- Designed and operated Docker-based infrastructure supporting multi-stage technical exercises for **50+ participants**.
- Coordinated technical implementation and infrastructure delivery across **4 project milestones** using Agile sprint workflows.

AI Instructor & Technical Coordinator – ADNOC / ADEK Jan 2025
Abu Dhabi, UAE

- Delivered applied AI/ML programs across **3 labs and 200+ systems**, achieving **90%+ participant completion**.
- Designed **15+ hands-on technical challenges** covering machine learning, optimization, and constrained problem-solving.
- Supported technical coordination between instructional requirements, lab infrastructure, and participant execution.

Independent Open-Source Security Researcher Jan 2019 – Present
GitHub / Codeberg *UAE*

Selected Engineering Projects

EMX Network Twin – Logistics Decision Platform | *Python, FastAPI, OR-Tools, SimPy, React, TypeScript* Aug 2026

- Built an end-to-end logistics digital twin for **3 UAE delivery networks and 22 facilities**, combining backend services, optimization, simulation, AI-assisted workflows, and a user-facing application.
- Developed a human-reviewed LLM interface that transforms natural-language planning requests into structured scenarios while deterministic engines calculate operational KPIs.
- Implemented **OR-Tools CP-SAT optimization** for fleet allocation, staffing, zones, and facilities under weekly cost and SLA constraints.
- Built seeded **SimPy and Monte Carlo simulations** with 50 replications, confidence intervals, baseline comparisons, and SQLite-backed audit records.

FEDROS OMNI – Agentic AI Platform | *LangGraph, LangChain, FAISS, Local LLMs* Jun 2026

- Built an end-to-end self-improving AI agent system using LangChain/LangGraph, FAISS, local LLMs, evaluation pipelines, and application-level observability.

- Improved benchmark task success by **35%** across 50+ tasks and reduced hallucination rate by **28%** through iterative evaluation and tool-use refinement.
- Implemented real-time multi-agent tracing, Arabic/RTL support, and KPI dashboards using metrics from **1,050+ interactions across 276 customers**.
- Ran **5,000+ autonomous agent iterations** on an NVIDIA A100, reducing experimental cost by approximately **40%**.

Containerized Full-Stack Platform | *Docker, Kubernetes, Nginx, CI/CD* Sep 2025

- Designed and deployed a containerized full-stack environment with SSL, RBAC, Nginx, automated CI/CD, and Kubernetes orchestration.
- Scaled a real-time multiplayer backend to support **100+ concurrent users** while maintaining >**99.9% uptime**.

Employee Identity Recognition System | *Python, OpenCV, FaceNet, ArcFace* Jun 2025

- Built an end-to-end computer vision pipeline achieving **94%+ macro accuracy** across 50+ identities.
- Improved performance under real-world lighting and occlusion conditions by tuning ArcFace embedding thresholds, reducing false acceptance rate by **30%**.

Passport Photo Compliance Automation | *TensorFlow, Computer Vision* Apr 2025

- Built a CNN-based compliance classifier across five ICAO photo criteria, achieving **96% F1-score** on held-out test data.
- Automated inference workflows that reduced estimated manual review workload by approximately **80%**.

Education

BSc, Software Development & AI Applications Expected 2026
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- AI/ML Club Board Member; lead workshops and develop technical exercises covering AI, machine learning, and AI security.
- Relevant study includes software engineering, systems programming, distributed data processing, Transformers, reinforcement learning, Spark, and Hadoop.

Cybersecurity Bootcamp 2024
Nomu Al Ghurair *Abu Dhabi, UAE*

Certifications & Technical Activities

- **Cisco Network Security** – 2025 | **Google Cybersecurity** – 2024
- **Kaggle ML Competitions:** Top-quartile finishes through end-to-end model development and feature engineering.
- **Facial Recognition Hackathon:** First-round finalist with a CNN-based identity classification system.
- **Cybersecurity Challenges:** Dubai CTF, Hack The Box, picoCTF, OverTheWire, and Pwnable.kr.